

Exam Date & Time: 16-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Industrial Nanotechnology [PHY_4344]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Compare the optical and electronic properties of bulk semiconductors with those of their nanostructured counterparts. (3)
 - A)
 - B) Differentiate the top-down and bottom-up approaches used in the synthesis of nanomaterials with examples. (3)
 - C) Discover the role of BioMEMS in personalized medicine and health monitoring. (4)
- 2) Identify the mechanism of action of photosensitizers in photodynamic therapy. (3)
 - A)
 - B) Find the concept of targeted drug delivery and its relevance in modern therapy. (3)
 - C) Discover the four main objectives and advantages of controlled drug delivery. (4)
- 3) Quantum dots are generally found to be more advantageous when compared to conventional fluorophores. Justify the statement with suitable reasons. (3)
 - A)
 - B) Assess how zeolites can be applied for wastewater treatment. What specific properties make them suitable for removing heavy metal ions. (3)
 - C) Analyze why nanocatalysts exhibit higher catalytic activity compared to their bulk counterparts. Give examples from industrial or environmental applications. (4)
- 4) Identify how precision irrigation systems contribute to sustainable water use in agriculture. (3)
 - A)
 - B) Assess how smart delivery systems can be used to supply nutrients to plants in a controlled and sustained manner. (3)

- C) Analyze the future of food processing with reference to personalized nutrition, 3D food printing, and lab-grown foods. (4)
- 5) Assess how combining polymers with inorganic nanoparticles can produce multifunctional nanofibers for specific applications. (3)
- A)
- B) Identify emerging trends such as 3D bioprinting, nanorobotics, and smart nanoscaffolds in advancing tissue engineering. (3)
- C) Analyze the application of silver or zinc oxide nanoparticles in producing antimicrobial or odour-resistant fabrics. (4)

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Event Name		Subject	
058-END SEMSTER-ODD-2025		PHY_4344:Industrial Nanotechnology	
#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* NA - CO Not Selected

Event Name		Subject	
058-END SEMSTER-ODD-2025		PHY_4344:Industrial Nanotechnology	
#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	10	30(60%)
4	Analysis	5	20(40%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)